

CHRISTOPHER J. EGAN

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Robotics Engineering and Computer Science junior at Worcester Polytechnic Institute specializing in computer vision, state estimation and machine learning. Experience in developing perception and control systems across research, embedded systems and over three years of international rocketry and robotics competitions

SKILLS & TECHNOLOGIES

Languages: C, C++, Python, Java, MATLAB, TypeScript, HTML, CSS, Racket

Software/Frameworks: Linux, Ubuntu, ROS, ROS2, Autodesk Inventor, OpenCV, CUDA, PyTorch, gcc/gcc++, Simulink, PlatformIO, git, Nerfstudio, Claude Code, React

Hardware: STM32H7, ESP32, Raspberry Pi, romi32U4, Realsense, Vicon, NVIDIA Jetson, LiDAR

EDUCATION

Worcester Polytechnic Institute (WPI)

2024 - 2028

B.S. Robotics Engineering & B.S. Computer Science (Dual Degree)

- Notable Courses: Object Oriented Design, Differential Equations, Algorithms, Operating Systems
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EXPERIENCE

Perception & Machine Learning Engineer

Feb. 2026 - Present

PeAR (Perception and Autonomous Robotics Lab)

Worcester, MA

- Researched and implemented optimal ways to get high quality, metric Gaussian splats fusing different combinations of tools such as Vicon, LiDAR, RGB images, depth images and IMUs
- Designed a pipeline to gather IMU, Vicon, RGB and depth data to render a metric Gaussian splat and generate a Euclidean Signed Distance Field for collision detection & path planning
- Contributed to an award finalist capstone project by integrating the ESDF pipeline with a dual-quaternion NMPC to guide a quadcopter.
- Integrated Gaussian splat and ESDF into reinforcement learning platform to generate photo realistic synthetic RGB, depth and event camera images as observations for drone obstacle avoidance.

Research Engineer

Aug. 2026 - Present

FuTURE with Boston Scientific

Worcester, MA

- Selected to contribute to development of AI assistant for endoscopy and stomach 3D reconstruction and volume estimation in collaboration with Boston Scientific.
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Leadership & Projects

Guidance, Navigation & Control Lead

Mar. 2026 - Aug. 2026

WPI High Powered Rocketry Club

Aug. 2025 - Present

- Developed a quaternion-based multiplicative extended Kalman filter (QM-EKF) that fused dual IMUs, GPS, barometer, and magnetometer measurements to estimate full 6-DoF vehicle state for autonomous rocket guidance in a MATLAB/Simulink flight simulation
- Ported QM-EKF to C++ on an STM32H7 microcontroller for a planned supersonic flight with an anticipated apogee of 10,000 feet, using canards for stabilization with poses from the QM-EKF
- Developed software to interface with two IMUs on an STM32H7 and collect 7 hours of synchronized data and run an Allan variance test for the QM-EKF's noise matrix